



CANE WALLET

Bender Alawwad, Blaise Driscoll, Preston
Gmernicki

01

Concept

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Our proposed solution is for a student to be able to add their cane card to their apple wallet and add updated scanners around campus that can use that system. Through their apple wallet, a student can view and manage their dining dollars, they can use it to scan into their dorm rooms and classroom at the University of Miami, they can more easily check books out from the library, and they can rent equipment from the gym more easily. This would increase the accessibility of everything on campus and make it easier for students to not lose or misplace their student IDs.

Why?

Currently a physical cane card is necessary and if someone's cane card is forgotten, or lost, it is difficult to get into buildings or do anything else on campus. This would be very helpful for students and faculty because it would make it easier and more efficient to do just about everything on campus.

02

Functional Requirements

Dining Dollars

- Be able to pay with dining dollars using your cane card via the apple wallet
- Students must have an active cane card with an apple wallet compatible device, and have enough money in their account to make the purchase
- If they do not have an active cane card, or there is not a sufficient balance to make the purchase, the system will decline the purchase
- Priority Rating : 1

Access to Dorms, Classrooms, Gym, etc

- We would integrate the cane card into Apple wallets to allow students to access their classrooms, dorms, gym, and more all in one.
- Student must have an active Cane Card and an Apple Wallet-compatible device, as well as permission to specific areas.
- Bring head of the iPhone to the sensor, in order to read information and allow access.
- Priority Rating : 0

Checking Out Resources

- Students would be able to use their cane card from their phones in order to check out resources from school facilities
- Students must have an active cane card with an apple wallet compatible device and have permission to use the specific resource
- The user will select the cane card in their apple wallet before tapping at an NFC scanner to check out the resource.
- Priority Rating : 2

03

Non Functional Requirements

Performance

- How fast and efficient the digital Cane Card works on the operating systems.
- Transactions (e.g., scanning for entry or purchasing food) should be processed within 2 seconds under normal operating conditions.
- Priority Rating: 0
- This is applicable to the use of Dining Dollars, Access to Dorms, Classrooms, and the Gym, and Checking out resources.

Usability

- Ease of adding and accessing the cane card in the apple wallet
- The cane card must be able to be added to the apple wallet simply with minimal steps, and accessed easily with a double tap. It also should allow the user to easily view their remaining dining dollar balance.
- Priority Rating: 2
- This is applicable to the use of Dining Dollars, Access to Dorms, Classrooms, and the Gym, and Checking out resources.

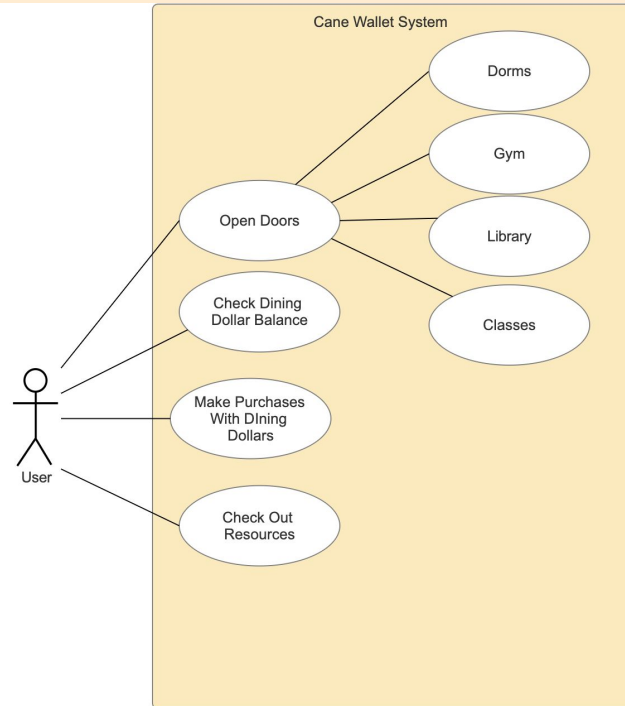
Compatibility

- The application should be compatible with iOS devices running iOS 14 or later, as well as allow for physical Cane Card usage as well
- The Cane Card should be able to work on all iOS devices iOS 14 or later, as well as allow for people to still be able to use their physical Cane Cards.
- Priority Rating: 0
- This is applicable to the use of Dining Dollars, Access to Dorms, Classrooms, and the Gym, and Checking out resources.

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Use Cases

Use Case Diagram



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System Constraints

Tool Constraints

Apple Wallet API

- The system must be compatible with Apple Wallet APIs
- Priority Rating: 0

Language Constraints

Programming Language

- The software must be developed Swift for Apple Wallet integration
- Priority Rating: 1

Platform Constraints

iOS Only

- This will only be available on iOS devices
- Priority Rating: 2

Hardware Constraints

NFC Scanner Compatibility

- All campus scanners must support NFC technology.
- Priority Rating: 0

Network Constraints

Wifi or Cellular Not Needed for Use

- No wifi is needed to use the Cane Card at systems. However, if it is not set up on your device, then you need wifi to add it to your device.
- Priority Rating: 3

Deployment Constraints

Incremental Development

- Having it be accessible from some areas of the campus first to allow for testing
- Priority Rating: 3

Transition & Support Constraints

Physical Cane Card Still Accessible by System

- If a student, for any reason, still needed to use their physical Cane Card, they should still have access to use their Dining Dollars and access into classrooms, dorms, and the Gym.
- Priority Rating: 0

Budget & Schedule Constraints

Budget Limitations

- Budget should cover the costs of software development, hardware upgrades, and the cost to deploy the software
- Priority Rating: 0

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Evolutionary Requirements

Future Expansion to Android

- Although we are focusing development towards iOS users, future versions should begin to add support for Android users
- The system must support multiple digital wallets
- Android NFC scanners must be tested for compatibility
- This expands the scope of our system to include everyone who uses an Android device
- Priority: 4

Improved System Performance

- There should always be effort to make the amount of time to scan from your Apple Wallet less
- Future updates should aim to reduce transaction time to less than 1 second
- This is applicable to the use of Dining Dollars, Access to Dorms, Classrooms, and the Gym, and Checking out resources.
- Priority Rating: 4